

MD. MIFTAHUR RAHMAN SWAPNIL

Product Designer & Builder | AI, Frontend & Automation

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SUMMARY

Final-year Computer Science undergraduate and product designer & builder dedicated to turning real-world friction into responsive digital products. Experienced across modern frontend engineering (Next.js, TypeScript), intuitive UI/UX prototyping, and AI systems integration. Creator of Edu51Portal, currently supporting ~100 active university peers with \$0 cloud infrastructure overhead. Undergraduate researcher with an accepted paper at IEEE OMLET 2026 on knowledge-graph RAG and ongoing work in EEG assistive interfaces. Driven by high craft, product empathy, and engineering resilient solutions from scratch.

SKILLS

- Product & Design:** UI/UX Design, Digital Product Design, Wireframing, High-Fidelity Prototyping (Figma), Design Systems, User Research
- Frontend Engineering:** Next.js 14/15 (App Router), React, TypeScript, Tailwind CSS, JavaScript (ES6+), HTML5/CSS3, Responsive UI
- Backend & Database:** Supabase (PostgreSQL, Row Level Security, Auth), Node.js, REST APIs, Google Drive API, Firebase
- AI & Research:** Knowledge Graphs & RAG Systems, Relational GCN, Neo4j, PyTorch, Deep Learning for EEG, Python
- Tools & Automation:** Git & GitHub, n8n Automation, Vercel, Docker (Basic), VS Code, LaTeX, Overleaf

FEATURED PROJECTS

- Edu51Portal** — Next.js 14, TypeScript, Supabase PostgreSQL, Google Drive API, Tailwind CSS Live
- Designed and built an academic platform serving **~100 active BUBT students**, delivering instant routine lookups, notices, and course materials.
 - Integrated Google Drive API with Supabase to distribute 1,000+ academic PDFs securely with **\$0 cloud storage infrastructure overhead**.
 - Engineered section-specific schedule filtering, fuzzy course search, and mobile-first responsive architecture.
- OpusGen AI** — Next.js, TypeScript, Generative Diffusion Models, REST APIs, Vercel Live
- Built an AI creative marketing suite providing automated studio lighting, background replacement, and 4x neural upscaling for e-commerce merchant catalogs.
 - Constructed batch processing export pipelines and multi-ratio ad canvas expansion, accelerating creative workflows from hours to under 30 seconds.
- Stark-OS Command Portfolio** — Next.js 14, TypeScript, Tailwind CSS, Supabase Telemetry, Vercel Live
- Engineered an interactive personal operating system portfolio (mrswapnil.me) with HUD audio feedback, multi-theme presets, and 98+ Lighthouse performance score.

RESEARCH & PUBLICATIONS

- Relation-Aware Graph Retrieval over a Curriculum Knowledge Graph for Prerequisite QA** Accepted (IEEE OMLET 2026)
- IEEE OMLET 2026** (Nairobi, Kenya) — Accepted with Minor Revision (Paper ID: 1017, Scheduled for IEEE Xplore & Scopus).
 - Authors: Md. Jahidul Kamal Islam, **Md. Miftahur Rahman Swapnil**, Md. Asif Ali, Shrabani Das, Shefayatuj Johara Chowdhury (BUBT CSE).
 - Engineered a curriculum knowledge graph RAG system with a 2-layer R-GCN encoder and DistMult decoder over **418 nodes and 558 typed edges** in Neo4j, achieving **45.5% exact-set match** and 100% correct abstention rate.
- AI-Enabled Smart Classroom Monitoring and Safety Automation** Completed (BUBT CSE)
- Engineered an IoT smart classroom safety and environmental monitoring system integrating sensor telemetry (DHT11, PIR, Ultrasonic, Sound) with automated door actuators and alert buzzers.
- EEG-Based Motor Imagery Classification** Ongoing / Running (2026)
- Investigating deep learning architectures for assistive neuro-rehabilitation and paralysis motor control interfaces.

EDUCATION

- Bachelor of Science in Computer Science & Engineering (CSE)** CGPA: 3.60 / 4.00
Bangladesh University of Business and Technology (BUBT) | Intake 51 2022 – 2026 (Expected)
- Higher Secondary Certificate (HSC) — Science** GPA: 5.00 / 5.00
Shaheed Police Smrity College 2021
- Secondary School Certificate (SSC) — Science** GPA: 5.00 / 5.00
Kadirabad BL High School, Pirganj, Rangpur 2019

LEADERSHIP & COMMUNITY

- BASIS Students' Forum — BUBT Chapter** — Member, Graphics Designer & Media and Publication Secretary 2023 – 2026
- Directed creative visual branding, digital promotional media, and publication outreach for tech seminars, workshops, and industry networking events.
- BUBT IT Club** — General Member 2022 – Present
- Internal University Event Management** — Active Coordinator & Event Organizer 2022 – Present

LANGUAGES

- Bangla** — Native / First Language
- English** — Professional Working Proficiency